

1. PRODUCT NAME

Fulvestrant-AFT 250 mg/5 mL (50 mg/mL) solution for injection in pre-filled syringe

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each pre-filled syringe contains 250 mg fulvestrant in 5 mL solution.

For the full list of excipients see section 6.1.

3. PHARMACEUTICAL FORM

Solution for injection in pre-filled syringe.

Clear, colourless to yellow, viscous liquid.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Fulvestrant-AFT is indicated:

- as monotherapy for the treatment of estrogen receptor positive, locally advanced or metastatic breast cancer in postmenopausal women:
 - who are human epidermal growth factor receptor 2 (HER2)-negative and not previously treated with endocrine therapy.
 - with disease relapse on or after adjuvant endocrine therapy, or disease progression on endocrine therapy.
- in combination with ribociclib for the treatment of hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative advanced or metastatic breast cancer in postmenopausal women, as initial endocrine based therapy or following disease progression on endocrine therapy.*
- in combination with abemaciclib for the treatment of women with hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative advanced or metastatic breast cancer with disease progression following endocrine therapy. *
- in combination with palbociclib for the treatment of hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative locally advanced or metastatic breast cancer in women who have received prior endocrine therapy (see Pharmacodynamic properties).*

In pre- or perimenopausal women, the combination treatment with palbociclib or abemaciclib should be combined with a luteinizing hormone releasing hormone (LHRH) agonist.

** This indication is only applicable in markets where palbociclib, ribociclib, or abemaciclib are registered.*

4.2 Posology and method of administration

Solution for intramuscular injection

Posology

Adult females (including the elderly).

The recommended dose is 500 mg at intervals of one month, with an additional 500 mg dose given two weeks after the initial dose.

When Fulvestrant-AFT is used in combination with palbociclib, abemaciclib or ribociclib, please also refer to the Summary of Product Characteristics of palbociclib, abemaciclib or ribociclib.

Prior to the start of treatment with the combination of Fulvestrant-AFT plus palbociclib or abemaciclib, and throughout its duration, pre/perimenopausal women should be treated with LHRH agonists according to local clinical practice.

Special populations

Paediatric patient

The safety and efficacy of Fulvestrant-AFT in children from birth to 18 years of age have not been established. Currently available data are described in sections Pharmacodynamic Properties and Pharmacokinetic Properties, but no recommendation on a posology can be made.

Renal impairment

No dose adjustments are recommended for patients with mild to moderate renal impairment (creatinine clearance ≥ 30 ml/min). Safety and efficacy have not been evaluated in patients with severe renal impairment (creatinine clearance < 30 ml/min) and, therefore, caution is recommended in these patients (see Special warnings and precautions for use).

Hepatic impairment

No dose adjustments are recommended for patients with mild to moderate hepatic impairment. However, as fulvestrant exposure may be increased, Fulvestrant-AFT should be used with caution in these patients. There are no data in patients with severe hepatic impairment (see Contraindications, Special warnings and precautions for use and Pharmacokinetic properties).

Method of administration

Fulvestrant-AFT should be administered as two consecutive 5 ml injections by slow intramuscular injection (1-2 minutes/injection), one in each buttock (gluteal area).

Caution should be taken if injecting Fulvestrant-AFT at the dorsogluteal site due to the proximity of the underlying sciatic nerve.

For detailed instructions for administration, see Instructions for administration and Special precautions for disposal

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.
- Pregnancy and lactation (see Pregnancy and lactation).
- Severe hepatic impairment (see Special warnings and precautions for use and Pharmacokinetic properties).

4.4 Special warnings and precautions for use

Fulvestrant-AFT should be used with caution in patients with mild to moderate hepatic impairment (see Posology and method of administration, Contraindications, and Pharmacokinetic Properties).

Fulvestrant-AFT should be used with caution in patients with severe renal impairment (creatinine clearance less than 30 ml/min) (see Pharmacokinetic properties).

Due to the intramuscular route of administration, Fulvestrant-AFT should be used with caution if treating patients with bleeding diatheses, thrombocytopenia or those taking anticoagulant treatment.

Thromboembolic events are commonly observed in women with advanced breast cancer and have been observed in clinical studies with Fulvestrant-AFT (see Undesirable effects). This should be taken into consideration when prescribing Fulvestrant-AFT to patients at risk.

Injection site related events including sciatica, neuralgia, neuropathic pain, and peripheral neuropathy have been reported with Fulvestrant-AFT injection. Caution should be taken while administering Fulvestrant-AFT at the dorsogluteal injection site due to the proximity of the underlying sciatic nerve (see Posology and method of administration and Undesirable effects).

There are no long-term data on the effect of fulvestrant on bone. Due to the mechanism of action of fulvestrant, there is a potential risk of osteoporosis.

The efficacy and safety of Fulvestrant-AFT (either as monotherapy or in combination with palbociclib) have not been studied in patients with critical visceral disease.

When Fulvestrant-AFT is combined with palbociclib, please also refer to the Summary of Product Characteristics of palbociclib.

Interference with estradiol antibody assays

Due to the structural similarity of fulvestrant and estradiol, fulvestrant may interfere with antibody based-estradiol assays and may result in falsely increased levels of estradiol.

Paediatric population

Fulvestrant-AFT is not recommended for use in children and adolescents as safety and efficacy have not been established in this group of patients (see Pharmacodynamic properties).

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates

4.5 Interaction with other medicines and other forms of interaction

A clinical interaction study with midazolam (substrate of CYP 3A4) demonstrated that fulvestrant does not inhibit CYP 3A4. Clinical interaction studies with rifampicin (inducer of CYP 3A4) and ketoconazole (inhibitor of CYP 3A4) showed no clinically relevant change in fulvestrant clearance. Dose adjustment is therefore not necessary in patients who are receiving fulvestrant and CYP 3A4 inhibitors or inducers concomitantly.

4.6 Fertility, pregnancy and lactation

Women of childbearing potential

Patients of child-bearing potential should use effective contraception during treatment with Fulvestrant-AFT and for 2 years after the last dose.

Pregnancy

Fulvestrant-AFT is contraindicated in pregnancy (see Contraindications). Fulvestrant has been shown to cross the placenta after single intramuscular doses in rat and rabbit. Studies in animals have shown

reproductive toxicity including an increased incidence of foetal abnormalities and deaths. If pregnancy occurs while taking Fulvestrant-AFT, the patient must be informed of the potential hazard to the foetus and potential risk for loss of pregnancy.

Breast-feeding

Breast-feeding must be discontinued during treatment with Fulvestrant-AFT. Fulvestrant is excreted in milk in lactating rats. It is not known whether fulvestrant is excreted in human milk. Considering the potential for serious adverse reactions due to fulvestrant in breast-fed infants, use during lactation is contraindicated (see Contraindications).

Fertility

The effects of Fulvestrant-AFT on fertility in humans has not been studied.

4.7 Effects on ability to drive and use machines

Fulvestrant-AFT has no or negligible influence on the ability to drive or use machines. However, since asthenia has been reported very commonly with Fulvestrant-AFT, caution should be observed by those patients who experience this adverse reaction when driving or operating machinery.

4.8 Undesirable effects

Summary of the safety profile

Monotherapy

This section provides information based on all adverse reactions from clinical studies, post-marketing studies or spontaneous reports. In the pooled dataset of fulvestrant monotherapy, the most frequently reported adverse reactions were injection site reactions, asthenia, nausea, and increased hepatic enzymes (ALT, AST, ALP).

In Table 1, the following frequency categories for adverse drug reactions (ADRs) were calculated based on the fulvestrant 500 mg treatment group in pooled safety analyses of studies that compared Fulvestrant 500 mg with fulvestrant 250 mg [CONFIRM (Study D6997C00002), FINDER 1 (Study D6997C00004), FINDER 2 (Study D6997C00006), and NEWEST (Study D6997C00003) studies], or from FALCON (Study D699BC00001) alone that compared fulvestrant 500 mg with anastrozole 1 mg. Where frequencies differ between the pooled safety analysis and FALCON, the highest frequency is presented. The frequencies in Table 1 were based on all reported events, regardless of the investigator assessment of causality. The median duration of fulvestrant 500 mg treatment across the pooled dataset (including the studies mentioned above plus FALCON) was 6.5 months.

Tabulated list of adverse reactions

Adverse reactions listed below are classified according to frequency and System Organ Class (SOC). Frequency groupings are defined according to the following convention: Very common ($\geq 1/10$), Common ($\geq 1/100$ to $< 1/10$), Uncommon ($\geq 1/1,000$ to $< 1/100$). Within each frequency grouping adverse reactions are reported in order of decreasing seriousness.

Table 1: Summary of ADRs seen in clinical studies for fulvestrant 500 mg

Adverse reactions by system organ class and frequency		
Infections and infestations	Common	Urinary tract infections
Blood and lymphatic system disorders	Common	Reduced platelet count ^e

Immune system disorders	Very common	Hypersensitivity reactions ^e
	Uncommon	Anaphylactic reactions
Metabolism and nutrition disorders	Common	Anorexia ^a
Nervous system disorders	Common	Headache
Vascular disorders	Very common	Hot flushes ^e
	Common	Venous thromboembolism ^a
Gastrointestinal disorders	Very common	Nausea
	Common	Vomiting, diarrhoea
Hepatobiliary disorders	Very common	Elevated hepatic enzymes (ALT, AST, ALP) ^a
	Common	Elevated bilirubin ^a
	Uncommon	Hepatic failure ^{c, f} , hepatitis ^f , elevated gamma-GT ^f
Skin and subcutaneous tissue disorders	Very common	Rash ^e
Musculoskeletal and connective tissue disorders	Very common	Joint and musculoskeletal pain ^d
	Common	Back pain ^a
Reproductive system and breast disorders	Common	Vaginal haemorrhage ^e
	Uncommon	Vaginal moniliasis ^f , leukorrhea ^f
General disorders and administration	Very common	Asthenia ^a , injection site reactions ^b
site conditions	Common	Neuropathy peripheral ^e , sciatica ^e
	Uncommon	Injection site haemorrhage ^f , injection site haematoma ^f , neuralgia ^{c, f}

a Includes adverse drug reactions for which the exact contribution of fulvestrant cannot be assessed due to the underlying disease.

b The term injection site reactions does not include the terms injection site haemorrhage, injection site haematoma, sciatica, neuralgia and neuropathy peripheral.

c The event was not observed in major clinical studies (CONFIRM, FINDER 1, FINDER 2, NEWEST). The frequency has been calculated using the upper limit of the 95% confidence interval for the point estimate. This is calculated as 3/560 (where 560 is the number of patients in the major clinical studies), which equates to a frequency category of 'uncommon'.

d Includes: arthralgia, and less frequently musculoskeletal pain, myalgia and pain in extremity.

e Frequency category differs between pooled safety dataset and FALCON.

f ADR was not observed in FALCON.

Description of selected adverse reactions

The descriptions included below are based on the safety analysis set of 228 patients who received at least one (1) dose of fulvestrant and 232 patients who received at least one (1) dose of anastrozole, respectively in the Phase 3 FALCON study.

Joint and musculoskeletal pain

In the FALCON study, the number of patients who reported an adverse reaction of joint and musculoskeletal pain was 65 (31.2%) and 48 (24.1%) for fulvestrant and anastrozole arms,

respectively. Of the 65 patients in the fulvestrant arm, 40% (26/65) of patients reported joint and musculoskeletal pain within the first month of treatment, and 66.2% (43/65) of patients within the first 3 months of treatment. No patients reported events that were CTCAE Grade ≥ 3 or that required a dose reduction, dose interruption, or discontinued treatment due to these adverse reactions.

Combination therapy with palbociclib

The overall safety profile of fulvestrant when used in combination with palbociclib is based on data from 517 patients with HR-positive, HER2-negative advanced or metastatic breast cancer in the randomised PALOMA3 study (see Pharmacodynamic properties). The most common ($\geq 20\%$) adverse reactions of any grade reported in patients receiving fulvestrant in combination with palbociclib were neutropenia, leukopenia, infections, fatigue, nausea, anaemia, stomatitis, diarrhoea, thrombocytopenia and vomiting. The most common ($\geq 2\%$) Grade ≥ 3 adverse reactions were neutropenia, leukopenia, infections, anaemia, AST increased, thrombocytopenia, and fatigue.

Table 2 reports the adverse reactions from PALOMA3. Median duration of exposure to fulvestrant was 11.2 months in the fulvestrant + palbociclib arm and 4.8 months in the fulvestrant + placebo arm. Median duration of exposure to palbociclib in the fulvestrant + palbociclib arm was 10.8 months.

Table 2: Adverse reactions from PALOMA3 (N=517)

System Organ Class Frequency Preferred Term ^a	Fulvestrant + Palbociclib (N=345)		Fulvestrant + placebo (N=172)	
	All Grades n (%)	Grade ≥ 3 n (%)	All Grades n (%)	Grade ≥ 3 n (%)
Infections and infestations				
<i>Very common</i>				
Infections ^b	188 (54.5)	19 (5.5)	60 (34.9)	6 (3.5)
Blood and lymphatic system disorders				
<i>Very common</i>				
Neutropenia ^c	290 (84.1)	240 (69.6)	6 (3.5)	0
Leukopenia ^d	207 (60.0)	132 (38.3)	9 (5.2)	1 (0.6)
Anaemia ^e	109 (31.6)	15 (4.3)	24 (14.0)	4 (2.3)
Thrombocytopenia ^f	88 (25.5)	10 (2.9)	0	0
<i>Uncommon</i>				
Febrile neutropenia	3 (0.9)	3 (0.9)	0	0
Metabolism and nutrition disorders				
<i>Very common</i>				
Decreased appetite	60 (17.4)	4 (1.2)	18 (10.5)	1 (0.6)
Nervous system disorders				
<i>Common</i>				
Dysgeusia	27 (7.8)	0	6 (3.5)	0
Eye disorders				
<i>Common</i>				

Lacrimation increased	25 (7.2)	0	2 (1.2)	0
Vision blurred	24 (7.0)	0	3 (1.7)	0
Dry eye	15 (4.3)	0	3 (1.7)	0
Respiratory, thoracic and mediastinal disorders				
<i>Common</i>				
Epistaxis	25 (7.2)	0	4 (2.3)	0
Gastrointestinal disorders				
<i>Very common</i>				
Nausea	124 (35.9)	2 (0.6)	53 (30.8)	1 (0.6)
Stomatitis ^g	104 (30.1)	3 (0.9)	24 (14.0)	0
Diarrhoea	94 (27.2)	0	35 (20.3)	2 (1.2)
Vomiting	75 (21.7)	2 (0.6)	28 (16.3)	1 (0.6)
Skin and subcutaneous tissue disorders				
<i>Very common</i>				
Alopecia	67 (19.4)	NA	11 (6.4)	NA
Rash ^h	63 (18.3)	3 (0.9)	10 (5.8)	0
<i>Common</i>				
Dry skin	28 (8.1)	0	3 (1.7)	0
General disorders and administration site conditions				
<i>Very common</i>				
Fatigue	152 (44.1)	9 (2.6)	54 (31.4)	2 (1.2)
Pyrexia	47 (13.6)	1 (0.3)	10 (5.8)	0
<i>Common</i>				
Asthenia	27 (7.8)	0	13 (7.6)	2 (1.2)
Investigations				
<i>Very Common</i>				
AST increased	40 (11.6)	11 (3.2)	13 (7.6)	4 (2.3)
<i>Common</i>				
ALT increased	30 (8.7)	7 (2.0)	10 (5.8)	1 (0.6)

ALT=alanine aminotransferase; AST=aspartate aminotransferase; N/n=number of patients; NA=Not applicable

a Preferred Terms (PTs) are listed according to MedDRA 17.1.

b Infections includes all PTs that are part of the System Organ Class Infections and infestations.

c Neutropenia includes the following PTs: Neutropenia, Neutrophil count decreased.

d Leukopenia includes the following PTs: Leukopenia, White blood cell count decreased.

e Anaemia includes the following PTs: Anaemia, Haemoglobin decreased, Haematocrit decreased.

f Thrombocytopenia includes the following PTs: Thrombocytopenia, Platelet count decreased.

g Stomatitis includes the following PTs: Aphthous stomatitis, Cheilitis, Glossitis, Glossodynia, Mouth ulceration, Mucosal inflammation, Oral pain, Oropharyngeal discomfort, Oropharyngeal pain, Stomatitis.

h Rash includes the following PTs: Rash, Rash maculo-papular, Rash pruritic, Rash erythematous, Rash papular, Dermatitis, Dermatitis acneiform, Toxic skin eruption.

Neutropenia

In patients receiving fulvestrant in combination with palbociclib in the PALOMA3 study, neutropenia of any grade was reported in 290 (84.1%) patients, with Grade 3 neutropenia being reported in 200

(58.0%) patients, and Grade 4 neutropenia being reported in 40 (11.6%) patients. In the fulvestrant + placebo arm (n=172), neutropenia of any grade was reported in 7 (4.1%) patients. There were no reports of Grade 3 and 4 neutropenia in the fulvestrant + placebo arm.

In patients receiving fulvestrant in combination with palbociclib, the median time to first episode of any grade neutropenia was 15 days (range: 13-512 days) and the median duration of Grade ≥ 3 neutropenia was 16 days. Febrile neutropenia has been reported in 3 (0.9%) patients receiving fulvestrant in combination with palbociclib.

Combination Therapy with Abemaciclib (MONARCH 2)

The safety of fulvestrant (500 mg) plus abemaciclib (150 mg twice daily) versus fulvestrant plus placebo was evaluated in MONARCH 2. The data described below reflect exposure to fulvestrant in 664 patients with HR-positive, HER2-negative advanced breast cancer who received at least one dose of Fulvestrant plus abemaciclib or placebo in MONARCH 2.

Median duration of treatment was 12 months for patients receiving fulvestrant plus abemaciclib and 8 months for patients receiving fulvestrant plus placebo.

Dose reductions due to an adverse reaction occurred in 43% of patients receiving fulvestrant plus abemaciclib. Adverse reactions leading to dose reductions $\geq 5\%$ of patients were diarrhea and neutropenia. Abemaciclib dose reduction due to diarrhea of any grade occurred in 19% of patients receiving fulvestrant plus abemaciclib compared to 0.4% of patients receiving fulvestrant plus placebo. Abemaciclib dose reductions due to neutropenia of any grade occurred in 10% of patients receiving fulvestrant plus abemaciclib compared to no patients receiving fulvestrant plus placebo.

Permanent study treatment discontinuation due to an adverse event was reported in 9% of patients receiving fulvestrant plus abemaciclib and in 3% of patients receiving fulvestrant plus placebo. Adverse reactions leading to permanent discontinuation for patients receiving fulvestrant plus abemaciclib were infection (2%), diarrhea (1%), hepatotoxicity (1%), fatigue (0.7%), nausea (0.2%), abdominal pain (0.2%), acute kidney injury (0.2%), and cerebral infarction (0.2%).

Deaths during treatment or during the 30-day follow up, regardless of causality, were reported in 18 cases (4%) of fulvestrant plus abemaciclib treated patients versus 10 cases (5%) of fulvestrant plus placebo treated patients. Causes of death for patients receiving fulvestrant plus abemaciclib included: 7 (2%) patient deaths due to underlying disease, 4 (0.9%) due to sepsis, 2 (0.5%) due to pneumonitis, 2 (0.5%) due to hepatotoxicity, and one (0.2%) due to cerebral infarction.

The most common adverse reactions reported ($\geq 20\%$) in the fulvestrant plus abemaciclib arm were diarrhea, fatigue, neutropenia, nausea, infections, abdominal pain, anemia, leukopenia, decreased appetite, vomiting, and headache (Table 3). The most frequently reported ($\geq 5\%$) Grade 3 or 4 adverse reactions were neutropenia, diarrhea, leukopenia, anemia, and infections.

Table 3: Adverse reactions $\geq 10\%$ of patients receiving Fulvestrant plus Abemaciclib and $\geq 2\%$ higher than Fulvestrant plus Placebo in MONARCH 2

Adverse Reactions	Fulvestrant plus Abemaciclib N=441			Fulvestrant plus Placebo N=223		
	All Grades %	Grade 3 %	Grade 4 %	All Grades %	Grade 3 %	Grade 4 %

Gastrointestinal Disorders						
Diarrhea	86	13	0	25	<1	0
Nausea	45	3	0	23	1	0
Abdominal pain ¹	35	2	0	16	1	0
Vomiting	26	<1	0	10	2	0
Stomatitis	15	<1	0	10	0	0
Infections and Infestations						
Infections ²	43	5	<1	25	3	<1
Blood and Lymphatic System Disorders						
Neutropenia ³	46	24	3	4	1	<1
Anemia ⁴	29	7	<1	4	1	0
Leukopenia ⁵	28	9	<1	2	0	0
Thrombocytopenia ⁶	16	2	1	3	0	<1
General Disorders and Administration Site Conditions						
Fatigue ⁷	46	3	0	32	<1	0
Edema peripheral	12	0	0	7	0	0
Pyrexia	11	<1	<1	6	<1	0
Metabolism and Nutrition Disorders						
Decreased appetite	27	1	0	12	<1	0
Respiratory, Thoracic and Mediastinal Disorders						
Cough	13	0	0	11	0	0
Skin and Subcutaneous Tissue Disorders						
Alopecia	16	0	0	2	0	0
Pruritus	13	0	0	6	0	0
Rash	11	1	0	4	0	0
Nervous System Disorders						
Headache	20	1	0	15	<1	0
Dysgeusia	18	0	0	3	0	0
Dizziness	12	1	0	6	0	0
Investigations						
Alanine aminotransferase increased	13	4	<1	5	2	0
Aspartate aminotransferase increased	12	2	0	7	3	0
Creatinine increased	12	<1	0	<1	0	0
Weight decreased	10	<1	0	2	<1	0

1. Includes abdominal pain, abdominal pain upper, abdominal pain lower, abdominal discomfort, abdominal tenderness.

2. Includes upper respiratory tract infection, urinary tract infection, lung infection, pharyngitis, conjunctivitis, sinusitis, vaginal infection, sepsis.
3. Includes neutropenia, neutrophil count decreased.
4. Includes anemia, hematocrit decreased, hemoglobin decreased, red blood cell count decreased.
5. Includes leukopenia, white blood cell count decreased.
6. Includes platelet count decreased, thrombocytopenia.
7. Includes asthenia, fatigue.

Additional adverse reactions in MONARCH 2 include venous thromboembolic events (deep vein thrombosis, pulmonary embolism, cerebral venous sinus thrombosis, subclavian vein thrombosis, axillary vein thrombosis, and DVT inferior vena cava), which were reported in 5% of patients treated with Fulvestrant plus abemaciclib as compared to 0.9% of patients treated with fulvestrant plus placebo.

Table 4: Laboratory abnormalities $\geq 10\%$ in patients receiving Fulvestrant plus Abemaciclib and $\geq 2\%$ higher than Fulvestrant plus Placebo in MONARCH 2

Laboratory Parameters	Fulvestrant plus Abemaciclib N=441			Fulvestrant plus Placebo N=223		
	All Grades %	Grade 3 %	Grade 4 %	All Grades %	Grade 3 %	Grade 4 %
Creatinine increased	98	1	0	74	0	0
White blood cell decreased	90	23	<1	33	<1	0
Neutrophil count decreased	87	29	4	30	4	<1
Anemia	84	3	0	33	<1	0
Lymphocyte count decreased	63	12	<1	32	2	0
Platelet count decreased	53	<1	1	15	0	0
Alanine aminotransferase increased	41	4	<1	32	1	0
Aspartate aminotransferase increased	37	4	0	25	4	<1

Combination Therapy with Ribociclib (MONALEESA-3)

The safety of Fulvestrant 500 mg plus ribociclib 600 mg versus fulvestrant plus placebo was evaluated in MONALEESA-3. The data described below reflect exposure to fulvestrant plus ribociclib in 483 out of 724 postmenopausal patients with HR-positive, HER2-negative advanced or metastatic breast cancer for initial endocrine based therapy or after disease progression on endocrine therapy who received at least one dose of fulvestrant plus ribociclib or placebo in MONALEESA-3. Median duration of treatment was 15.8 months for fulvestrant plus ribociclib and 12 months for fulvestrant plus placebo.

Dose reductions due to adverse reactions occurred in 32% of patients receiving fulvestrant plus ribociclib and in 3% of patients receiving fulvestrant plus placebo. Among patients receiving fulvestrant plus ribociclib, 8% were reported to have permanently discontinued both Fulvestrant plus ribociclib, and 9% were reported to have discontinued ribociclib alone due to ARs. Among patients

receiving fulvestrant plus placebo, 4% were reported to have permanently discontinued both fulvestrant and placebo and 2% were reported to have discontinued placebo alone due to ARs.

Adverse reactions leading to treatment discontinuation of fulvestrant plus ribociclib (as compared to fulvestrant plus placebo) were ALT increased (5% vs. 0%), AST increased (3% vs. 0.6%), and vomiting (1% vs. 0%).

The most common adverse reactions (reported at a frequency $\geq 20\%$ on the fulvestrant plus ribociclib arm and $\geq 2\%$ higher than fulvestrant plus placebo) were neutropenia, infections, leukopenia, cough, nausea, diarrhea, vomiting, constipation, pruritus, and rash. The most frequently reported Grade 3/4 adverse reactions (reported at a frequency $\geq 5\%$) in patients receiving fulvestrant plus ribociclib in descending frequency were neutropenia, leukopenia, infections, and abnormal liver function tests.

Adverse reactions and laboratory abnormalities occurring in patients in MONALEESA-3 are listed in Table 5 and Table 6, respectively.

Table 5: Adverse reactions occurring in $\geq 10\%$ and $\geq 2\%$ higher than Fulvestrant plus Placebo arm in MONALEESA-3 (All Grades)

Adverse Reactions	Fulvestrant plus Ribociclib N=483			Fulvestrant plus Placebo N=241		
	All Grades %	Grade 3 %	Grade 4 %	All Grades %	Grade 3 %	Grade 4 %
Infections and Infestations						
Infections ¹	42	5	0	30	2	0
Blood and Lymphatic System Disorders						
Neutropenia	69	46	7	2	0	0
Leukopenia	27	12	<1	<1	0	0
Anemia	17	3	0	5	2	0
Metabolism and Nutrition Disorders						
Decreased appetite	16	<1	0	13	0	0
Nervous System Disorders						
Dizziness	13	<1	0	8	0	0
Respiratory, Thoracic and Mediastinal Disorders						
Cough	22	0	0	15	0	0
Dyspnea	15	1	<1	12	2	0
Gastrointestinal Disorders						
Nausea	45	1	0	28	<1	0
Diarrhea	29	<1	0	20	<1	0
Vomiting	27	1	0	13	0	0
Constipation	25	<1	0	12	0	0
Abdominal pain	17	1	0	13	<1	0
Skin and Subcutaneous Tissue Disorders						
Alopecia	19	0	0	5	0	0
Pruritus	20	<1	0	7	0	0
Rash	23	<1	0	7	0	0
General Disorders and Administration Site Conditions						
Edema peripheral	15	0	0	7	0	0
Pyrexia	11	<1	0	7	0	0
Investigations						
Alanine aminotransferase increased	15	7	2	5	<1	0
Aspartate aminotransferase increased	13	5	1	5	<1	0

Grading according to CTCAE 4.03.

CTCAE=Common Terminology Criteria for Adverse Events; N=number of patients

1. Infections; urinary tract infections; respiratory tract infections; gastroenteritis; sepsis (<1%).

Additional adverse reactions in MONALEESA-3 for patients receiving Fulvestrant plus ribociclib included asthenia (14%), dyspepsia (10%), thrombocytopenia (9%), dry skin (8%), dysgeusia (7%),

electrocardiogram QT prolonged (6%), dry mouth (5%), vertigo (5%), dry eye (5%), lacrimation increased (4%), erythema (4%), hypocalcemia (4%), blood bilirubin increased (1%), and syncope (1%).

Table 6: Laboratory abnormalities occurring in ≥10% of patients in MONALEESA-3

Laboratory parameters	Fulvestrant plus Ribociclib N=483			Fulvestrant plus Placebo N=241		
	All Grades %	Grade 3 %	Grade 4 %	All Grades %	Grade 3 %	Grade 4 %
Hematology						
Leukocyte count decreased	95	25	<1	26	<1	0
Neutrophil count decreased	92	46	7	21	<1	0
Hemoglobin decreased	60	4	0	35	3	0
Lymphocyte count decreased	69	14	1	35	4	<1
Platelet count decreased	33	<1	1	11	0	0
Chemistry						
Creatinine increased	65	<1	<1	33	<1	0
Gamma-glutamyl transferase increased	52	6	1	49	8	2
Aspartate aminotransferase increased	49	5	2	43	3	0
Alanine aminotransferase increased	44	8	3	37	2	0
Glucose serum decreased	23	0	0	18	0	0
Phosphorous decreased	18	5	0	8	<1	0
Albumin decreased	12	0	0	8	0	0

Overdose

There are isolated reports of overdose with fulvestrant in humans. If overdose occurs, symptomatic supportive treatment is recommended. Animal studies suggest that no effects other than those related directly or indirectly to anti-estrogenic activity were evident with higher doses of fulvestrant.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antineoplastic antioestrogen, ATC Code: LO2BA03.

Mechanism of action and pharmacodynamic effects

Fulvestrant is a competitive estrogen receptor (ER) antagonist with an affinity comparable to estradiol. Fulvestrant blocks the trophic actions of estrogens without any partial agonist (estrogen-like) activity. The mechanism of action is associated with down-regulation of estrogen receptor (ER) protein levels. Clinical studies in postmenopausal women with primary breast cancer have shown that fulvestrant significantly down-regulates ER protein in ER positive tumours compared with placebo. There was also

a significant decrease in progesterone receptor expression consistent with a lack of intrinsic estrogen agonist effects. It has also been shown that fulvestrant 500 mg downregulates ER and the proliferation marker Ki67, to a greater degree than fulvestrant 250 mg in breast tumours in postmenopausal neoadjuvant setting.

Clinical safety and efficacy in advanced breast cancer

Monotherapy

A Phase 3 clinical study was completed in 736 postmenopausal women with advanced breast cancer who had disease recurrence on or after adjuvant endocrine therapy or progression following endocrine therapy for advanced disease. The study included 423 patients whose disease had recurred or progressed during anti-estrogen therapy (AE subgroup) and 313 patients whose disease had recurred or progressed during aromatase inhibitor therapy (AI subgroup). This study compared the efficacy and safety of Fulvestrant 500 mg (n=362) with Fulvestrant 250 mg (n=374). Progression-free survival (PFS) was the primary endpoint; key secondary efficacy endpoints included objective response rate (ORR), clinical benefit rate (CBR) and overall survival (OS).

Efficacy results for the CONFIRM study are summarized in Table 7.

Table 7: Summary of results of the primary efficacy endpoint (PFS) and key secondary efficacy endpoints in the CONFIRM study

Variable	Type of estimate; Treatment comparison	Fulvestrant 500 mg (N=362)	Fulvestrant 250 mg (N=374)	Comparison between groups (Fulvestrant 500 mg/ Fulvestrant 250 mg)		
				Hazard ratio	95% CI	p-value
PFS	K-M median in months; hazard ratio					
All Patients		6.5	5.5	0.80	0.68, 0.94	0.006
-AE subgroup (n=423)		8.6	5.8	0.76	0.62, 0.94	0.013
-AI subgroup (n=313) ^a		5.4	4.1	0.85	0.67, 1.08	0.195
Updated OS ^b	K-M median in months; hazard ratio					
All Patients		26.4	22.3	0.81	0.69, 0.96	0.016 ^C
-AE subgroup (n=423)		30.6	23.9	0.79	0.63, 0.99	0.038 ^C
-AI subgroup (n=313) ^a		24.1	20.8	0.86	0.67, 1.11	0.241 ^C
Variable	Type of estimate; Treatment comparison	Fulvestrant 500 mg (N=362)	Fulvestrant 250 mg (N=374)	Comparison between groups (Fulvestrant 500 mg/ Fulvestrant 250 mg)		
				Absolute difference in %	95% CI	
ORR ^d	% of patients with OR; absolute difference in %					
All Patients		13.8	14.6	-0.8	-5.8, 6.3	

-AE subgroup (n=296)	18.1	19.1	-1.0	-8.2, 9.3
-AI subgroup (n=205) ^a	7.3	8.3	-1.0	-5.5, 9.8
CBR ^e	% of patients with CB; absolute difference in %			
All Patients	45.6	39.6	1.28	-1.1, 13.3
-AE subgroup (n=423)	52.4	45.1	1.34	-2.2, 16.6
-AI subgroup (n=313) ^a	36.2	32.3	1.19	-6.1, 15.2

a Fulvestrant is indicated in patients whose disease had recurred or progressed on an anti-estrogen therapy. The results in the AI subgroup are inconclusive.

b OS is presented for the updated and final survival analyses at 75% maturity.

c Nominal p-value with no adjustments made for multiplicity between the initial overall survival analyses at 50% maturity and the updated survival analyses at 75% maturity.

d ORR was assessed in patients who were evaluable for response at baseline (ie., those with measurable disease at baseline: 240 patients in the fulvestrant 500 mg group and 261 patients in the fulvestrant 250 mg group).

e Patients with a best objective response of complete response, partial response or stable disease ≥ 24 weeks.

PFS: Progression-free survival; ORR: Objective response rate; OR: Objective response; CBR: Clinical benefit rate; CB: Clinical benefit; OS: Overall survival; K-M: Kaplan-Meier; CI: Confidence interval; AI: Aromatase inhibitor; AE: Anti-estrogen.

A Phase 3, randomised, double-blind, double-dummy, multicentre study of Fulvestrant 500 mg versus anastrozole 1 mg was conducted in postmenopausal women with ER-positive and/or PgR-positive locally advanced or metastatic breast cancer who had not previously been treated with any hormonal therapy. A total of 462 patients were randomised 1:1 sequentially to receive either fulvestrant 500 mg or anastrozole 1 mg.

Randomisation was stratified by disease setting (locally advanced or metastatic), prior chemotherapy for advanced disease, and measurable disease.

The primary efficacy endpoint of the study was investigator assessed progression-free survival (PFS) evaluated according to RECIST 1.1 (Response Evaluation Criteria in Solid Tumours). Key secondary efficacy endpoints included overall survival (OS) and objective response rate (ORR).

Patients enrolled in this study had a median age of 63 years (range 36-90). The majority of patients (87.0%) had metastatic disease at baseline. Fifty-five percent (55.0%) of patients had visceral metastasis at baseline. A total of 17.1% of patients received a prior chemotherapy regimen for advanced disease; 84.2% of patients had measurable disease.

Consistent results were observed across the majority of pre-specified patient subgroups. For the subgroup of patients with disease limited to non-visceral metastasis (n=208), the HR was 0.592 (95% CI: 0.419, 0.837) for the Fulvestrant arm compared to the anastrozole arm. For the subgroup of patients with visceral metastasis (n=254), the HR was 0.993 (95% CI: 0.740, 1.331) for the Fulvestrant arm compared to the anastrozole arm. The efficacy results of the FALCON study are presented in Table 8 and Figure 1.

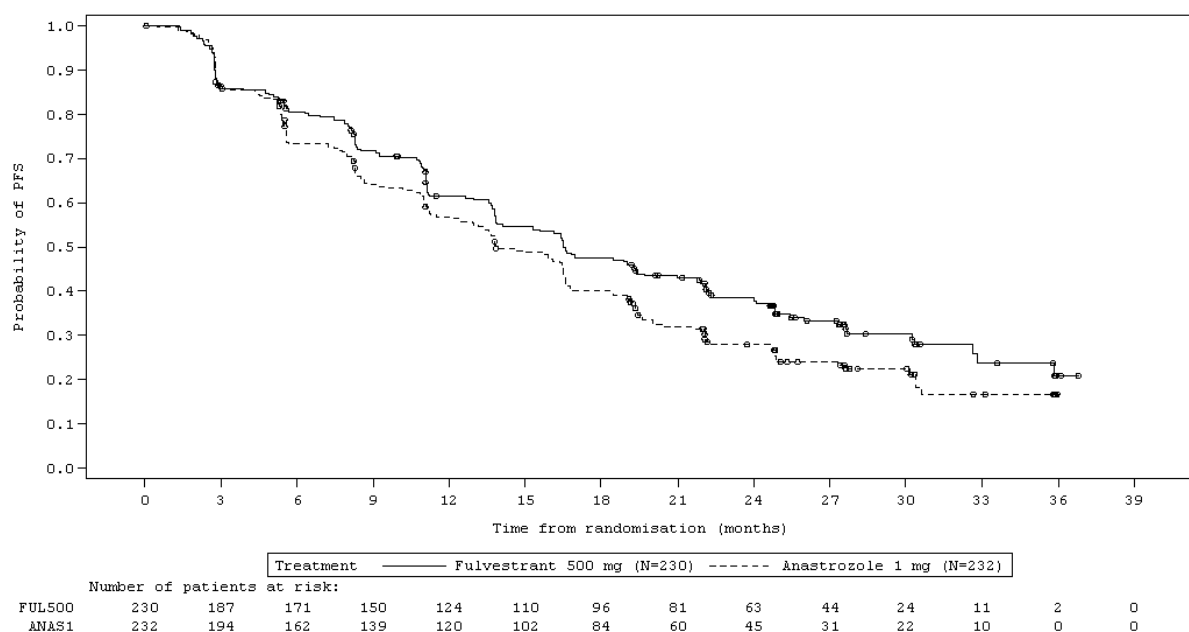
Table 8: Summary of results of the primary efficacy endpoint (PFS) and key secondary efficacy endpoints (Investigator Assessment, Intent-To-Treat Population) – FALCON study

	Fulvestrant 500mg (N=230)	Anastrozole 1 mg (N=232)
Progression-Free Survival		
Number of PFS Events (%)	143 (62.2%)	166 (71.6%)
PFS Hazard Ratio (95% CI) and p-value	HR 0.797 (0.637 - 0.999) p = 0.0486	
PFS Median [months (95% CI)]	16.6 (13.8, 21.0)	13.8 (12.0, 16.6)
Number of OS Events*	67 (29.1%)	75 (32.3%)
OS Hazard Ratio (95% CI) and p-value	HR 0.875 (0.629 – 1.217) p = 0.4277	
ORR**	89 (46.1%)	88 (44.9%)
ORR Odds Ratio (95% CI) and p-value	OR 1.074 (0.716 – 1.614) p = 0.7290	
Median DoR (months)	20.0	13.2
CBR	180 (78.3%)	172 (74.1%)
CBR Odds Ratio (95% CI) and p-value	OR 1.253 (0.815 – 1.932) p = 0.3045	

*(31% maturity)-not final OS analysis

**for patients with measurable disease

Figure 1: Kaplan-Meier Plot of Progression-Free Survival (Investigator Assessment, Intent-To-Treat Population) – FALCON Study



Two Phase 3 clinical studies were completed in a total of 851 postmenopausal women with advanced breast cancer who had disease recurrence on or after adjuvant endocrine therapy or progression following endocrine therapy for advanced disease. Seventy seven percent (77%) of the study population had estrogen receptor positive breast cancer. These studies compared the safety and

efficacy of monthly administration of Fulvestrant 250 mg versus the daily administration of 1 mg anastrozole (aromatase inhibitor). Overall, Fulvestrant at the 250 mg monthly dose was at least as effective as anastrozole in terms of progression-free survival, objective response, and time to death. There were no statistically significant differences in any of these endpoints between the two treatment groups. Progression-free survival was the primary endpoint. Combined analysis of both studies showed that 83% of patients who received Fulvestrant progressed, compared with 85% of patients who received anastrozole. Combined analysis of both studies showed the hazard ratio of Fulvestrant 250 mg to anastrozole for progression-free survival was 0.95 (95% CI 0.82 to 1.10). The objective response rate for Fulvestrant 250 mg was 19.2% compared with 16.5% for anastrozole. The median time to death was 27.4 months for patients treated with Fulvestrant and 27.6 months for patients treated with anastrozole. The hazard ratio of Fulvestrant 250 mg to anastrozole for time to death was 1.01 (95% CI 0.86 to 1.19).

Combination Therapy

Patients with HR-positive, HER2-negative advanced or metastatic breast cancer who have had disease progression on or after prior adjuvant or metastatic endocrine therapy

Fulvestrant 500mg in combination therapy with palbociclib (PALOMA-3)

A Phase 3, international, randomised, double-blind, parallel-group, multicentre study of Fulvestrant 500 mg plus palbociclib 125 mg versus Fulvestrant 500 mg plus placebo was conducted in women with HR-positive, HER2-negative locally advanced breast cancer not amenable to resection or radiation therapy with curative intent or metastatic breast cancer, regardless of their menopausal status, whose disease progressed after prior endocrine therapy in the (neo) adjuvant or metastatic setting.

A total of 521 pre/peri- and postmenopausal women who had progressed on or within 12 months from completion of adjuvant endocrine therapy on or within 1 month from prior endocrine therapy for advanced disease, were randomised 2:1 to Fulvestrant plus palbociclib or Fulvestrant plus placebo and stratified by documented sensitivity to prior hormonal therapy, menopausal status at study entry (pre/peri- versus postmenopausal), and presence of visceral metastases. Pre/perimenopausal women received the LHRH agonist goserelin. Patients with advanced/metastatic, symptomatic, visceral spread, that were at risk of life-threatening complications in the short term (including patients with massive uncontrolled effusions [pleural, pericardial, peritoneal], pulmonary lymphangitis, and over 50% liver involvement), were not eligible for enrolment into the study.

Patients continued to receive assigned treatment until objective disease progression, symptomatic deterioration, unacceptable toxicity, death, or withdrawal of consent, whichever occurred first. Crossover between treatment arms was not allowed.

Patients were well matched for baseline demographics and prognostic characteristics between the Fulvestrant plus palbociclib arm and the Fulvestrant plus placebo arm. The median age of patients enrolled in this study was 57 years (range 29, 88). In each treatment arm the majority of patients were White, had documented sensitivity to prior hormonal therapy, and were postmenopausal. Approximately 20% of patients were pre/perimenopausal. All patients had received prior systemic therapy and most patients in each treatment arm had received a previous chemotherapy regimen for their primary diagnosis. More than half (62%) had an ECOG PS of 0, 60% had visceral metastases, and 60% had received more than 1 prior hormonal regimen for their primary diagnosis.

The primary endpoint of the study was investigator-assessed PFS evaluated according to RECIST 1.1. Supportive PFS analyses were based on an Independent Central Radiology Review. Secondary endpoints included OR, CBR, overall survival (OS), safety, and time-to-deterioration (TTD) in pain endpoint.

The study met its primary endpoint of prolonging investigator-assessed PFS at the interim analysis conducted on 82% of the planned PFS events; the results crossed the pre-specified Haybittle-Peto efficacy boundary ($\alpha=0.00135$), demonstrating a statistically significant prolongation in PFS and a clinically meaningful treatment effect. A more mature update of efficacy data is reported in Table 9.

After a median follow-up time of 45 months, the final OS analysis was performed based on 310 events (60% of randomised patients). A 6.9-month difference in median OS in the palbociclib plus fulvestrant arm compared with the placebo plus fulvestrant arm was observed; this result was not statistically significant at the prespecified significance level of 0.0235 (1-sided). In the placebo plus fulvestrant arm, 15.5% of randomised patients received palbociclib and other CDK inhibitors as post-progression subsequent treatments.

The results from the investigator-assessed PFS and final OS data from PALOMA3 study are presented in Table 9. The relevant Kaplan-Meier plots are shown in Figures 2 and 3, respectively.

Table 9: Efficacy results – PALOMA3 study (Investigator assessment, intent-to-treat population)

	Updated analysis (23 October 2015 cut-off)	
	Fulvestrant plus Palbociclib (N=347)	Fulvestrant plus Placebo (N=174)
Progression-Free Survival		
Median [months (95% CI)]	11.2 (9.5, 12.9)	4.6 (3.5, 5.6)
Hazard ratio (95% CI) and p-value	0.497 (0.398, 0.620), p <0.000001	
Secondary end-points*		
OR [% (95% CI)]	26.2 (21.7, 31.2)	13.8 (9.0, 19.8)
OR (measurable disease) [% (95% CI)]	33.7 (28.1, 39.7)	17.4 (11.5, 24.8)
CBR [% (95% CI)]	68.0 (62.8, 72.9)	39.7 (32.3, 47.3)
Final overall survival (OS) (13 April 2018 cutoff)		
Number of events (%)	201 (57.9)	109 (62.6)
Median [months 95% CI]	34.9 (28.8, 40.0)	28.0 (23.6, 34.6)
Hazard ratio (95% CI) and p-value [†]	0.814 (0.644, 1.029) p=0.0429 ^{†*}	

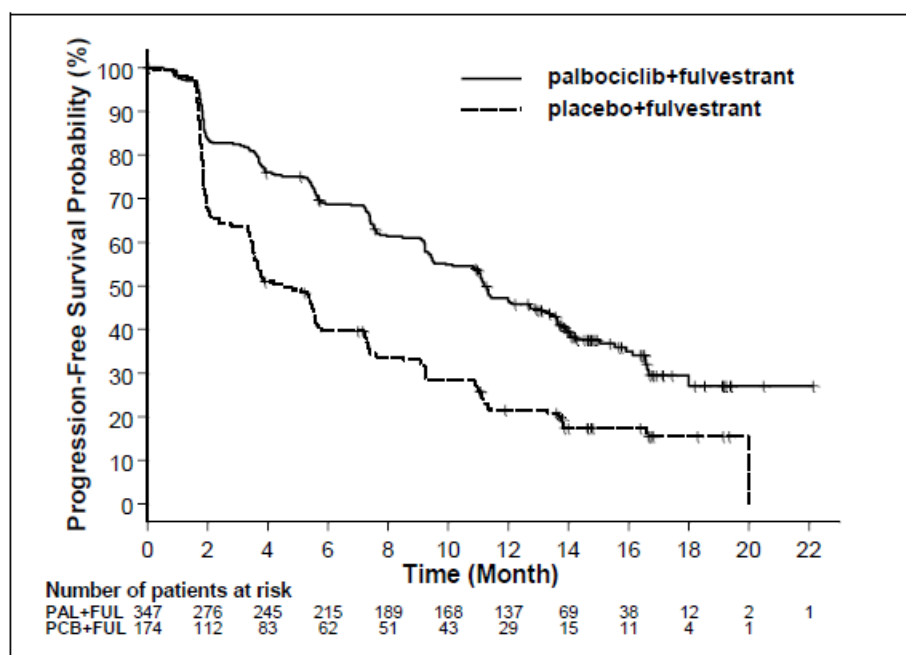
CBR=clinical benefit response; CI=confidence interval; N=number of patients; OR=objective response;

Secondary endpoint results are based on confirmed and unconfirmed responses according to RECIST 1.1.

* Not statistically significant.

† 1-sided p-value from the log-rank test stratified by the presence of visceral metastases and sensitivity to prior endocrine therapy per randomisation.

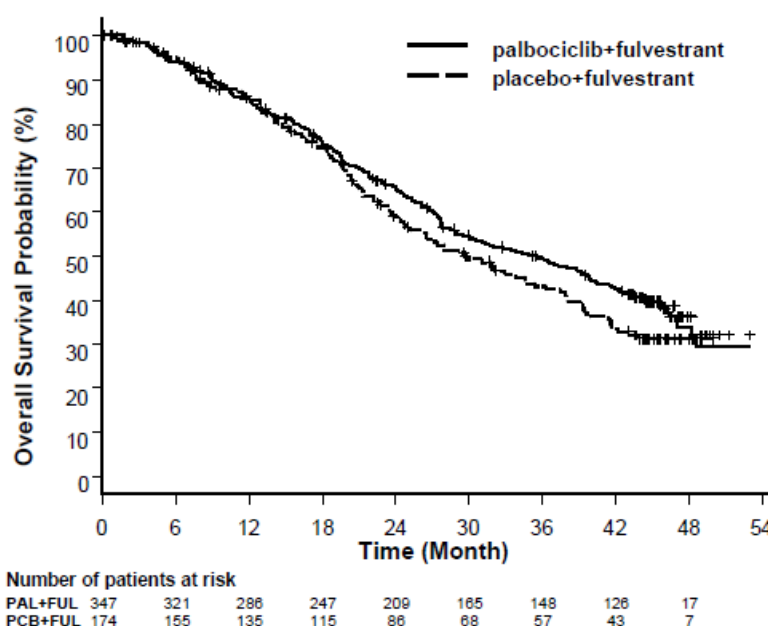
Figure 2: Kaplan-Meier plot of progression-free survival (investigator assessment, intent-to-treat population) – PALOMA3 study (23 October 2015 cutoff)



FUL=fulvestrant; PAL=palbociclib; PCB=placebo.

A reduction in the risk of disease progression or death in the Fulvestrant plus palbociclib arm was observed in all individual patient subgroups defined by stratification factors and baseline characteristics. This was evident for pre/perimenopausal women (HR of 0.46 [95% CI: 0.28, 0.75]) and postmenopausal women (HR of 0.52 [95% CI: 0.40, 0.66]) and patients with visceral site of metastatic disease (HR of 0.50 [95% CI: 0.38, 0.65]) and non-visceral site of metastatic disease (HR of 0.48 [95% CI: 0.33, 0.71]). Benefit was also observed regardless of lines of prior therapy in the metastatic setting, whether 0 (HR of 0.59 [95% CI: 0.37, 0.93]), 1 (HR of 0.46 [95% CI: 0.32, 0.64]), 2 (HR of 0.48 [95% CI: 0.30, 0.76]), or ≥ 3 lines (HR of 0.59 [95% CI: 0.28, 1.22]).

Figure 3: Kaplan-Meier plot of overall survival (intent-to-treat population) – PALOMA3 study (13 April 2018 cutoff)



FUL=fulvestrant; PAL=palbociclib; PCB=placebo

Additional efficacy measures (OR and TTR) assessed in the sub-groups of patients with or without visceral disease are displayed in Table 10.

Table 10: Efficacy results in visceral and non-visceral disease from PALOMA3 study (intent-to- treat population)

	Visceral Disease		Non-visceral Disease	
	Fulvestrant plus Palbociclib (N=206)	Fulvestrant plus Placebo (N=105)	Fulvestrant plus Palbociclib (N=141)	Fulvestrant plus Placebo (N=69)
OR [% (95% CI)]	35.0 (28.5, 41.9)	13.3 (7.5, 21.4)	13.5 (8.3, 20.2)	14.5 (7.2, 25.0)
TTR*, Median [months (range)]	3.8 (3.5, 16.7)	5.4 (3.5, 16.7)	3.7 (1.9, 13.7)	3.6 (3.4, 3.7)

*Response results based on confirmed and unconfirmed responses.

N=number of patients; CI=confidence interval; OR= objective response; TTR=time to first tumour response.

Patient-reported symptoms were assessed using the European Organization for Research and Treatment of Cancer (EORTC) quality of life questionnaire (QLQ)-C30 and its Breast Cancer Module (EORTC QLQ-BR23). A total of 335 patients in the Fulvestrant plus palbociclib arm and 166 patients in the Fulvestrant plus placebo arm completed the questionnaire at baseline and at least 1 post-baseline visit.

Time-to-Deterioration was pre-specified as time between baseline and first occurrence of ≥ 10 points increase from baseline in pain symptom scores. Addition of palbociclib to Fulvestrant resulted in a symptom benefit by significantly delaying Time-to-Deterioration in pain symptom compared with Fulvestrant plus placebo (median 8.0 months versus 2.8 months; HR of 0.64 [95% CI: 0.49, 0.85]; $p < 0.001$).

Fulvestrant 500 mg in combination with Abemaciclib 150 mg (MONARCH 2)

MONARCH 2 (NCT02107703) was a randomized, placebo-controlled, multi-center study conducted in women with HR-positive, HER2-negative metastatic breast cancer with disease progression following endocrine therapy treated with Fulvestrant plus abemaciclib versus Fulvestrant plus placebo. Randomization was stratified by disease site (visceral, bone only, or other) and by sensitivity to prior endocrine therapy (primary or secondary resistance). A total of 669 patients received intramuscular injection of Fulvestrant 500 mg on Days 1 and 15 of cycle 1 and then on Day 1 of cycle 2 and beyond (28-day cycles), plus abemaciclib or placebo orally twice daily. Pre/perimenopausal women were enrolled in the study and received the gonadotropin-releasing hormone agonist goserelin for at least 4 weeks prior to and for the duration of MONARCH 2. Patients remained on continuous treatment until development of progressive disease or unmanageable toxicity.

Patient median age was 60 years (range, 32-91 years), and 37% of patients were older than 65. The majority were White (56%), and 99% of patients had an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. Twenty percent (20%) of patients had de novo metastatic disease, 27% had bone only disease, and 56% had visceral disease. Twenty-five percent (25%) of patients had primary endocrine therapy resistance. Seventeen percent (17%) of patients were pre- or perimenopausal.

The efficacy results from the MONARCH 2 study are summarized in Table 11, Figure 4 and Figure 5. PFS assessment based on a blinded independent radiologic review was consistent with the investigator assessment. Consistent results were observed across patient stratification subgroups of disease site and endocrine therapy resistance for PFS and OS.

Table 11: Efficacy results in MONARCH 2 (intent-to-treat population)

	Fulvestrant plus Abemaciclib	Fulvestrant plus Placebo
Progression-Free Survival (Investigator Assessment)	N=446	N=223
Number of patients with an event (n, %)	222 (49.8)	157 (70.4)
Median (months, 95% CI)	16.4 (14.4, 19.3)	9.3 (7.4, 12.7)
Hazard ratio (95% CI) ¹	0.553 (0.449, 0.681)	
p-value	p<0.0001	
Overall Survival²		
Number of death (n, %)	211 (47.3)	127 (57.0)
Median OS in months (95% CI)	46.7 (39.2, 52.5)	37.3 (34.4, 43.2)
Hazard ratio (95% CI) ¹	0.757 (0.606, 0.945)	
p-value ¹	p=0.0137	
Objective Response for Patients with Measurable Disease	N=318	N=164
Objective response rate ³ (n, %)	153 (48.1)	35 (21.3)
95% CI	42.6, 53.6	15.1, 27.6

Abbreviations: CI=confidence interval, OS=overall survival.

1. Stratified by disease site (visceral metastases vs. bone-only metastases vs. other) and endocrine therapy resistance (primary resistance vs. secondary resistance)
2. Data from a pre-specified interim analysis (77% of the number of events needed for the planned final analysis) with the p-value compared with the allocated alpha of 0.021.
3. Complete response + partial response.

Figure 4: Kaplan-Meier Curves of Progression-Free Survival: Fulvestrant plus Abemaciclib versus Fulvestrant plus Placebo (MONARCH 2)

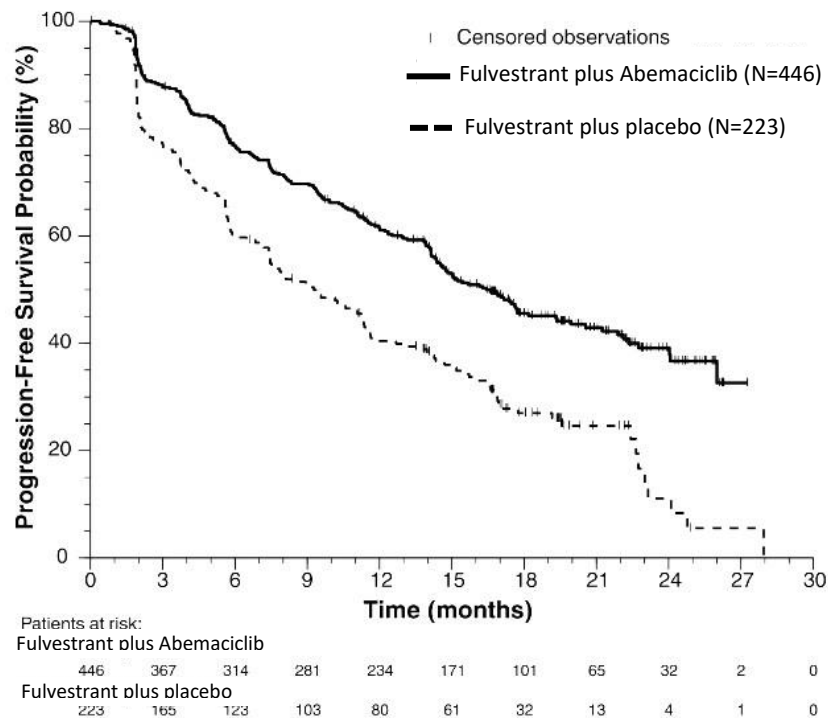
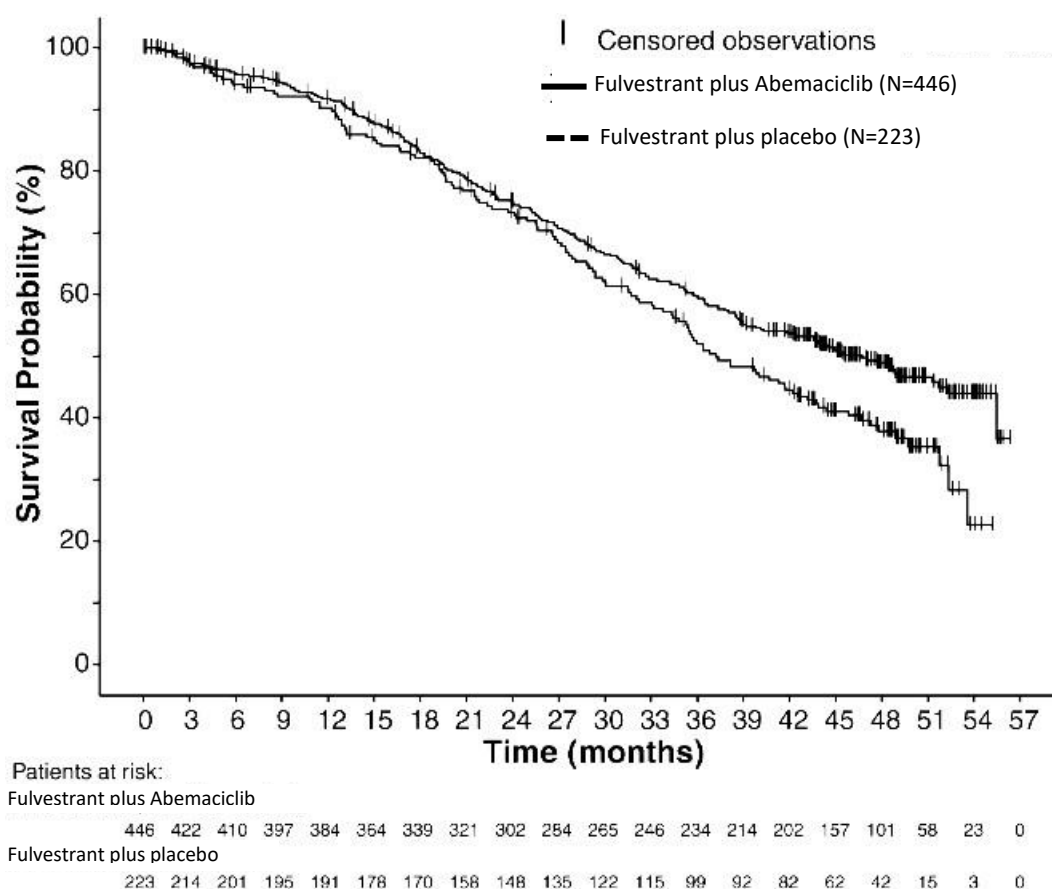


Figure 5: Kaplan-Meier curves of overall survival: Fulvestrant plus Abemaciclib versus Fulvestrant plus Placebo (MONARCH 2)



Postmenopausal women with HR-positive, HER2-negative advanced or metastatic breast cancer for initial endocrine based therapy or after disease progression on endocrine therapy

Fulvestrant 500 mg in combination with Ribociclib 600 mg (MONALEESA-3)

MONALEESA-3 (NCT 02422615) was a randomized double-blind, placebo-controlled study of fulvestrant plus ribociclib versus fulvestrant plus placebo conducted in postmenopausal women with hormone receptor positive, HER2-negative, advanced breast cancer who have received no or only one line of prior endocrine treatment.

A total of 726 patients were randomized in a 2:1 ratio to receive fulvestrant plus ribociclib or fulvestrant plus placebo and stratified according to the presence of liver and/or lung metastases and prior endocrine therapy for advanced or metastatic disease. Fulvestrant 500 mg was administered intramuscularly on Days 1, 15, 29, and once monthly thereafter, with either ribociclib 600 mg or placebo given orally once daily for 21 consecutive days followed by 7 days off until disease progression or unacceptable toxicity. The major efficacy outcome measure for the study was investigator-assessed progression-free survival (PFS) using Response Evaluation Criteria in Solid Tumors (RECIST) v1.1.

Patients enrolled in this study had a median age of 63 years (range 31 to 89). Of the patients enrolled, 47% were 65 years and older, including 14% age 75 years and older. The patients enrolled were primarily Caucasian (85%), Asian (9%), and Black (0.7%). Nearly all patients (99.7%) had an ECOG performance status of 0 or 1. First and second line patients were enrolled in this study (of which 19%

had de novo metastatic disease). Forty-three percent (43%) of patients had received chemotherapy in the adjuvant vs. 13% in the neoadjuvant setting and 59% had received endocrine therapy in the adjuvant vs. 1% in the neoadjuvant setting prior to study entry. Twenty-one percent (21%) of patients had bone only disease and 61% had visceral disease. Demographics and baseline disease characteristics were balanced and comparable between study arms.

The efficacy results from MONALEESA-3 are summarized in Table 12, Figure 6 and Figure 7. Consistent results were observed in stratification factor subgroups of disease site and prior endocrine treatment for advanced disease.

Table 12: Efficacy results – MONALEESA-3 (investigator assessment, intent-to-treat population)

	Fulvestrant plus Ribociclib	Fulvestrant plus Placebo
Progression-free survival*	N=484	N=242
Events (n, %)	210 (43.4%)	151 (62.4%)
Median (months, 95% CI)	20.5 (18.5, 23.5)	12.8 (10.9, 16.3)
Hazard Ratio (95% CI)	0.593 (0.480 to 0.732)	
p-value ¹	<0.0001	
Overall Survival	N=484	N=242
Events (n, %)	167 (34.5%)	108 (44.6%)
Median (months, 95% CI)	NR (42.5, NR)	40.0 (37.0, NR)
Hazard Ratio (95% CI)	0.724 (0.568, 0.924)	
p-value ¹	0.00455	
Overall Response Rate²	N=379	N=181
Patients with measurable disease (95% CI)	40.9 (35.9, 45.8)	28.7 (22.1, 35.3)

Abbreviation: NR, not reached

1. p-value is obtained from the one-sided log-rank

2. Based on confirmed responses

* Investigator Assessment

Figure 6: Kaplan-Meier progression free survival curves – MONALEESA-3 (intent-to-treat population, investigator assessment)

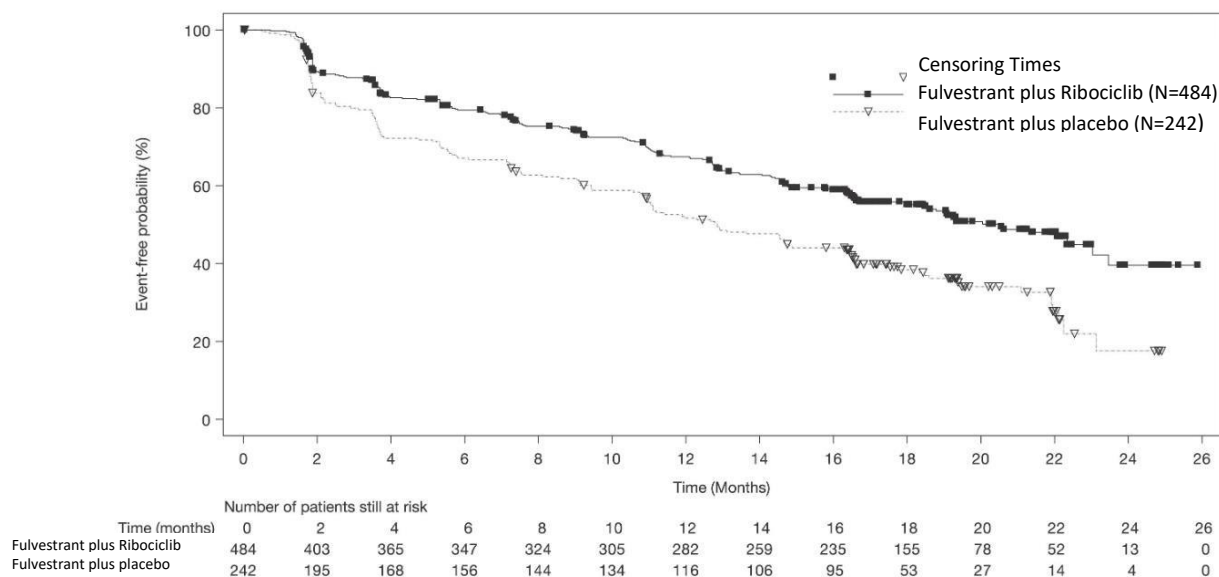
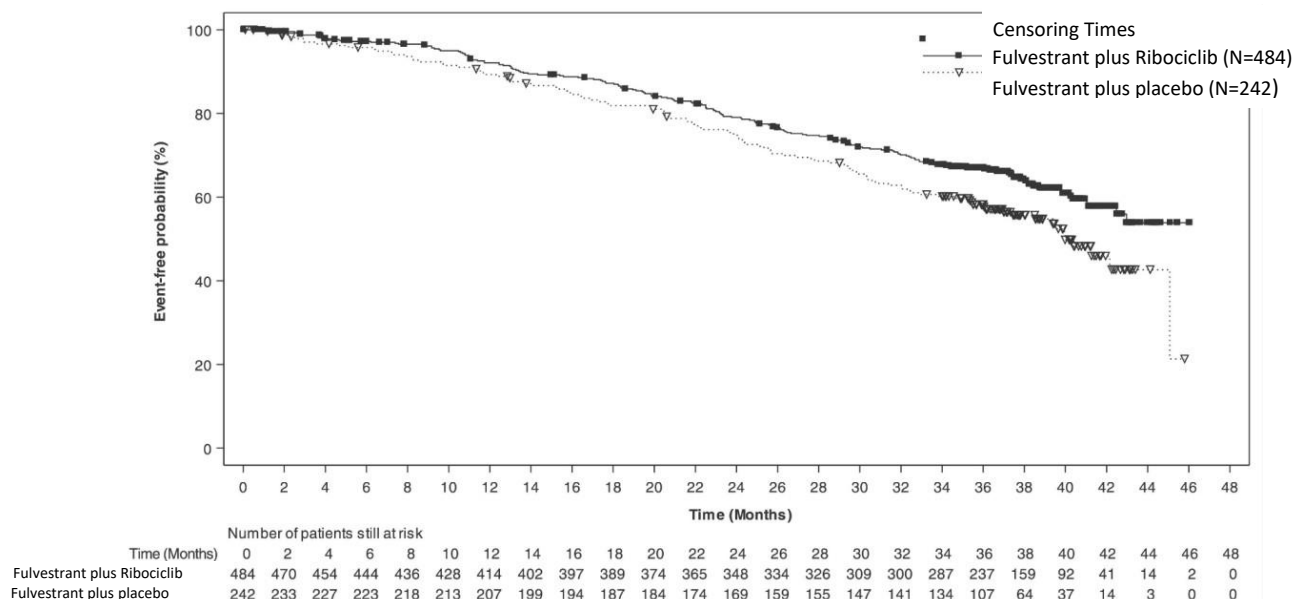


Figure 7: Kaplan-Meier plot of overall survival – MONALEESA-3 (intent-to-treat population)



Effects on the postmenopausal endometrium

Preclinical data do not suggest a stimulatory effect of fulvestrant on the postmenopausal endometrium. A 2-week study in healthy postmenopausal volunteers treated with 20 µg per day of ethinylestradiol showed that pre-treatment with fulvestrant 250 mg resulted in significantly reduced stimulation of the postmenopausal endometrium, compared to pre-treatment with placebo, as judged by ultrasound measurement of endometrium thickness.

Neoadjuvant treatment for up to 16 weeks in breast cancer patients treated with either fulvestrant 500 mg or fulvestrant 250 mg did not result in clinically significant changes in endometrial thickness, indicating a lack of agonist effect. There is no evidence of adverse endometrial effects in the breast cancer patients studied. No data are available regarding endometrial morphology.

In two short-term studies (1 and 12 weeks) in premenopausal patients with benign gynaecologic disease, no significant differences in endometrial thickness were observed by ultrasound measurement between fulvestrant and placebo groups.

Effects on bone

There are no long-term data on the effect of fulvestrant on bone. Neoadjuvant treatment for up to 16 weeks in breast cancer patients with either fulvestrant 500 mg or fulvestrant 250 mg did not result in clinically significant changes in serum bone-turnover markers.

Paediatric population

Fulvestrant is not indicated for use in children.

An open-label Phase 2 study investigated the safety, efficacy and pharmacokinetics of fulvestrant in 30 girls aged 1 to 8 years with Progressive Precocious Puberty associated with McCune Albright Syndrome (MAS). The paediatric patients received 4 mg/kg monthly intramuscular dose of fulvestrant. This 12-month study investigated a range of MAS endpoints and showed a reduction in the frequency of vaginal bleeding and a reduction in the rate of bone age advancement. The steady-state trough concentrations of fulvestrant in children in this study were consistent with that in adults (see section Pharmacokinetic properties). There were no new safety concerns arising from this small study, but 5-year data are yet not available.

5.2 Pharmacokinetic properties

Absorption

After administration of fulvestrant long-acting intramuscular injection, fulvestrant is slowly absorbed and maximum plasma concentrations ($C_{\max}$) are reached after about 5 days. Administration of fulvestrant 500 mg regimen achieves exposure levels at, or close to, steady state within the first month of dosing (mean [CV]: AUC 475 [33.4%] ng.days/ml, $C_{\max}$ 25.1 [35.3%] ng/ml, $C_{\min}$ 16.3 [25.9%] ng/ml, respectively). At steady state, fulvestrant plasma concentrations are maintained within a relatively narrow range with up to an approximately 3-fold difference between maximum and trough concentrations. After intramuscular administration, the exposure is approximately dose proportional in the dose range 50 to 500 mg.

Distribution

Fulvestrant is subject to extensive and rapid distribution. The large apparent volume of distribution at steady state ($V_{d_{ss}}$) of approximately 3 to 5 l/kg suggests that distribution is largely extravascular. Fulvestrant is highly (99%) bound to plasma proteins. Very low density lipoprotein (VLDL), low density lipoprotein (LDL), and high density lipoprotein (HDL) fractions are the major binding components. No interaction studies were conducted on competitive protein binding. The role of sex hormone-binding globulin (SHBG) has not been determined.

Biotransformation

The metabolism of fulvestrant has not been fully evaluated, but involves combinations of a number of possible biotransformation pathways analogous to those of endogenous steroids. Identified metabolites (includes 17-ketone, sulphone, 3-sulphate, 3- and 17-glucuronide metabolites) are either less active or exhibit similar activity to fulvestrant in anti-estrogen models. Studies using human liver preparations and recombinant human enzymes indicate that CYP 3A4 is the only P450 isoenzyme

involved in the oxidation of fulvestrant, however, non-P450 routes appear to be more predominant in vivo. In vitro data suggest that fulvestrant does not inhibit CYP450 isoenzymes.

Elimination

Fulvestrant is eliminated mainly in metabolised form. The major route of excretion is via the faeces, with less than 1% being excreted in the urine. Fulvestrant has a high clearance, 11 ± 1.7 ml/min/kg, suggesting a high hepatic extraction ratio. The terminal half-life ($t_{1/2}$) after intramuscular administration is governed by the absorption rate and was estimated to be 50 days.

Special Populations

In a population pharmacokinetic analysis of data from Phase 3 studies, no difference in fulvestrant's pharmacokinetic profile was detected with regard to age (range 33 to 89 years), weight (40-127 kg) or race.

Renal impairment

Mild to moderate impairment of renal function did not influence the pharmacokinetics of fulvestrant to any clinically relevant extent.

Hepatic impairment

The pharmacokinetics of fulvestrant has been evaluated in a single-dose clinical study conducted in women with mild to moderate hepatic impairment (Child-Pugh class A and B). A high dose of a shorter duration intramuscular injection formulation was used. There was up to about 2.5-fold increase in AUC in women with hepatic impairment compared to healthy subjects. In patients administered fulvestrant, an increase in exposure of this magnitude is expected to be well tolerated. Women with severe hepatic impairment (Child-Pugh class C) were not evaluated.

Paediatric population

The pharmacokinetics of fulvestrant has been evaluated in a clinical study conducted in 30 girls with Progressive Precocious Puberty associated with McCune Albright Syndrome (see section Pharmacodynamic properties). The paediatric patients were aged 1 to 8 years and received 4 mg/kg monthly intramuscular dose of fulvestrant. The geometric mean (standard deviation) steady state trough concentration ($C_{min,ss}$) and AUC_{ss} was 4.2 (0.9) ng/mL and 3680 (1020) ng*hr/mL, respectively. Although the data collected were limited, the steady-state trough concentrations of fulvestrant in children appear to be consistent with those in adults.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

- Alcohol (Ethanol) 96%
- Benzyl alcohol
- Benzyl benzoate
- Castor oil

6.2 Incompatibilities

In the absence of compatibility studies, this medicine must not be mixed with other medicines.

6.3 Shelf life

Please refer to expiry date on label/outer carton.

6.4 Special precautions for storage

Store at 2 °C to 8 °C (in a refrigerator). Do not freeze. Store in the original package in order to protect from light.

6.5 Nature and contents of container

Fulvestrant-AFT solution is packaged in a 5 mL clear neutral glass (Type I) barrel, stoppered by a bromobutyl plunger stopper, assembled with a plunger rod and a 5 mL backstop, and then presented in a tray with a safety needle (SafetyGlide™) for connection to the barrel. Two (2) syringes per box with leaflet.

6.6 Instructions for administration and special precautions for disposal and other handling.

Administer the injection according to the local guidelines for performing large volume intramuscular injections.

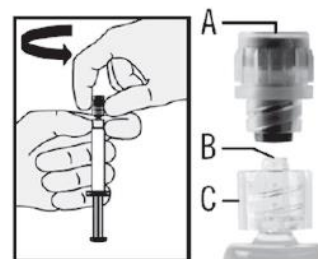
NOTE: Due to the proximity of the underlying sciatic nerve, caution should be taken if administering Fulvestrant-AFT at the dorsogluteal injection site (see Special warnings and precautions for use).

Warning - Do not autoclave safety needle (BD SafetyGlide™ Shielding Hypodermic Needle) before use. Hands must remain behind the needle at all times during use and disposal.

For each of the two syringes:

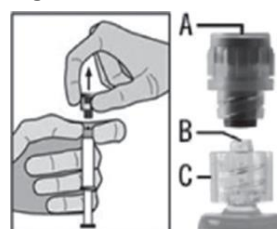
- Remove glass syringe barrel from tray and check that it is not damaged.
- Peel open the safety needle (SafetyGlide™) outer packaging.
- Inspect drug product in glass syringe for any visible particulate matter or discoloration prior to use. Discard if particulate matter or discoloration is present.
- Parenteral solutions must be inspected visually for particulate matter and discolouration prior to administration.
- Hold the syringe upright on the ribbed part (C). With the other hand, take hold of the cap (A) and carefully **twist the cap counter-clockwise** until the cap disconnect for removal (see Figure 8).

Figure 8



- Pull the cap (A) off in a straight upward direction. **Do not touch the sterile syringe tip** (Luer-Lok) (B) (see Figure 9).

Figure 9



- Attach the safety needle to the syringe tip (Luer-Lok) and twist until firmly seated (see Figure 10).
- Confirm that needle is locked to the Luer connector before moving or tilting the syringe out of the vertical plane to avoid spillage of syringe contents.
- Pull shield straight off needle to avoid damaging needlepoint.
- Remove needle sheath.
- Expel excess gas from the syringe
- Administer intramuscularly slowly (1-2 minutes/injection) into the buttock (gluteal area).
- For user convenience, the needle bevel-up position is oriented to the lever arm (see Figure 11).

Figure 10

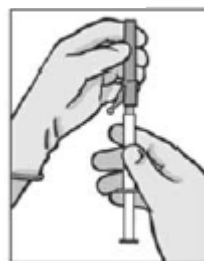


Figure 11

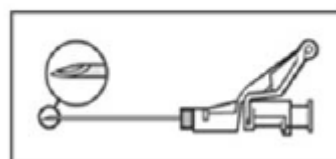


Figure 12



- After injection, immediately activate the lever arm to deploy the needle shielding by applying a single-finger stroke to the activation assisted lever arm to push the lever arm completely forward. Listen for a click. Confirm that the needle shielding has completely covered the needle (see Figure 12).
- NOTE: Activate away from self and others.

Pre-filled syringes are for single use only.

Discard used syringes into a sharps container.

Return unused and expired medicines to your local pharmacy for disposal.

7. MEDICINE SCHEDULE

Prescription Medicine

8. PRODUCT REGISTRATION HOLDER IN MALAYSIA

AFT Pharmaceuticals (S.E. Asia) SDN BHD
Suite 9.01-Regus, Level 9, Menara Summit
Persiaran Kewajipan, USJ 1, UEP
47600 Subang Jaya, Selangor, Malaysia

9. DATE OF REVISION OF THE TEXT

27 November 2024

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